

The Significance of Neural Networks in NLP



Our world is currently dominated by automation. Instant help from chatbots, email filters, voice assistants, predictive text, and search results are all examples of artificial intelligence (AI) and one of its primary branches, natural language processing (NLP). In this article we take a look at how these technologies use neural networks.

Speaking, writing, listening to, and recognising human language are the most fascinating and challenging things artificial intelligence (AI) can do. And now natural language processing (NLP) is set to revolutionise how AI understands human desires and behaviours, with deep learning being the game changer in this field.

Neurons in the human brain are interconnected to help make a judgement call; neural networks are inspired by these neurons and assist a computer to make distinct decisions or predictions. Neural networks are made up of a web of interconnected nodes, each of which is responsible for performing simple calculations to achieve the desired

outcomes. Some of the important neural networks are:

- Recurrent neural networks (RNN)
- Artificial neural networks (ANN)
- Convolutional neural networks (CNN)

Recurrent neural networks

Recurrent neural network (RNN) is a powerful deep learning technology that is crucial to understand if you want to get into machine learning. We are all probably indirectly using RNN based applications in our smartphones and websites. Speech recognition, language translation, and stock predictions all use RNNs. These neural networks excel at modelling sequence data.